

Variable Environments I

Today's Agenda:

- Quiz
- Variable Environments

Variable Environments

Recently we've been talking about variation in space

Variation in time is also super important!

Types of variation

- Variation around a mean (e.g., temperature, precipitation, tides, etc.)
- Long-term trends (e.g., climate change, desertification)

Key ideas

- Succession
 - Intermediate disturbance hypothesis
- Temporal variation and coexistence
- Alternative Stable States (a phrase that is rarely abbreviated)
- Slow change but rapid exposure to new conditions

Succession

Succession = pattern of community change following disturbance

Primary succession = starting from scratch

- E.g., newly formed islands, land scraped bare by glaciers

Secondary succession = starting from a large disturbance

- E.g., fires, agriculture, floods, etc.
- Soil, seeds, maybe some resilient organisms may remain

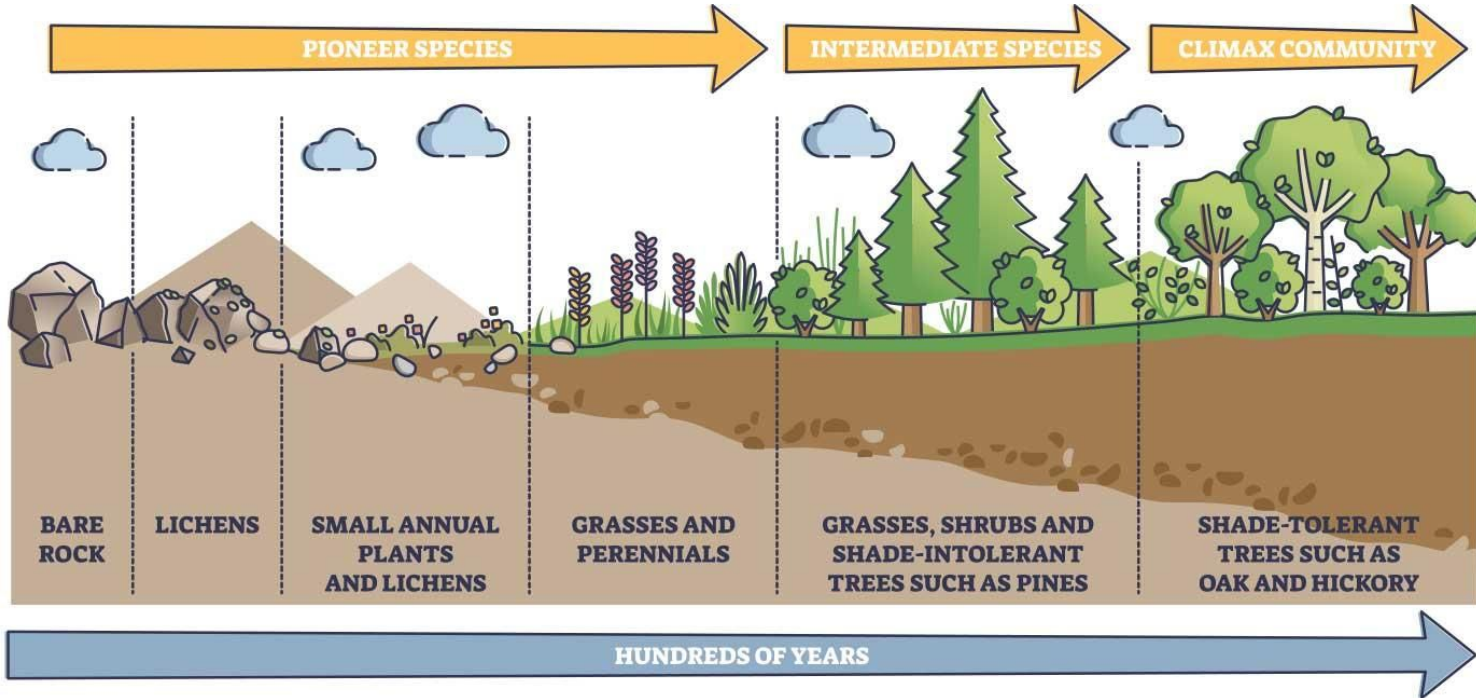
Succession

Some early ecologists thought succession would always end the same

- Any location, if disturbed, would always return to the same **climax community**

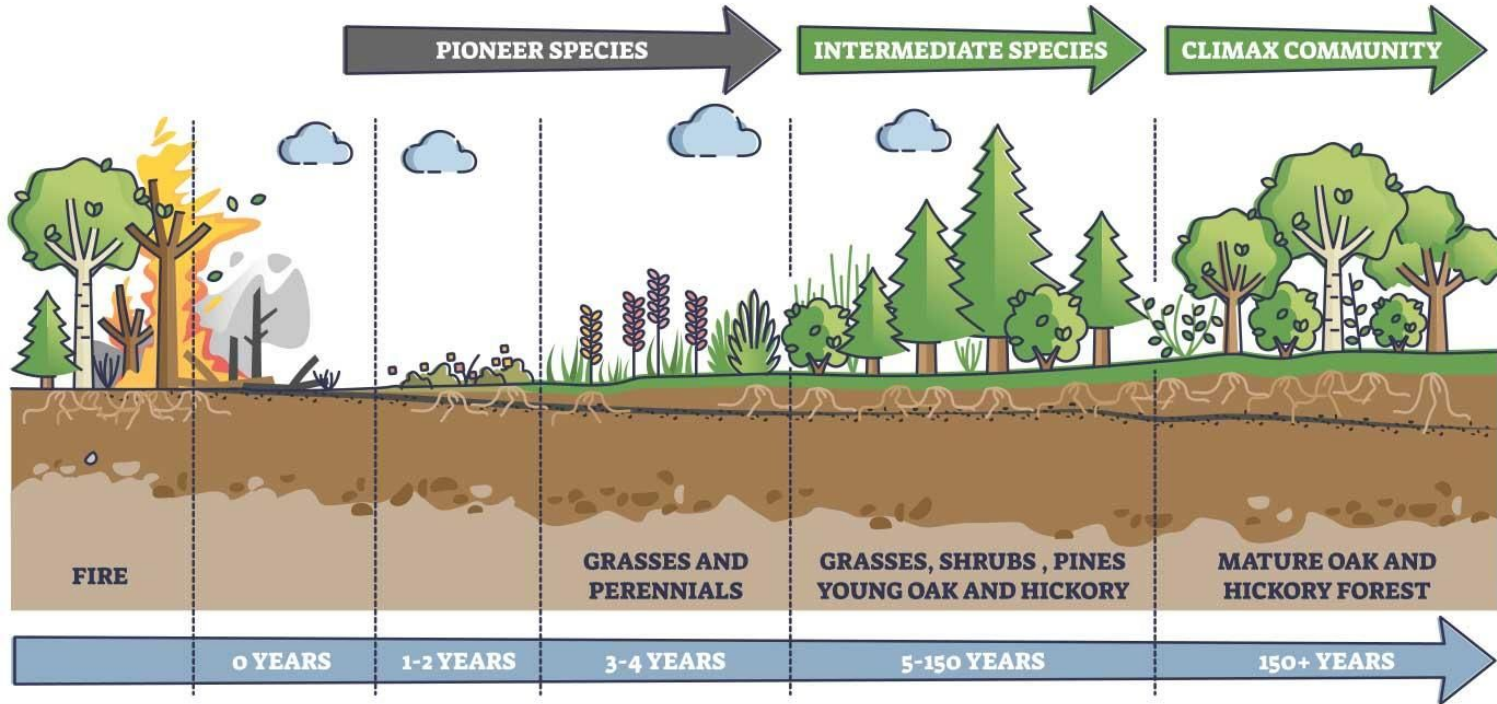
Succession

PRIMARY SUCCESSION



Succession

SECONDARY SUCCESSION



Small-scale succession

Note that these dynamics can occur on different spatial scales

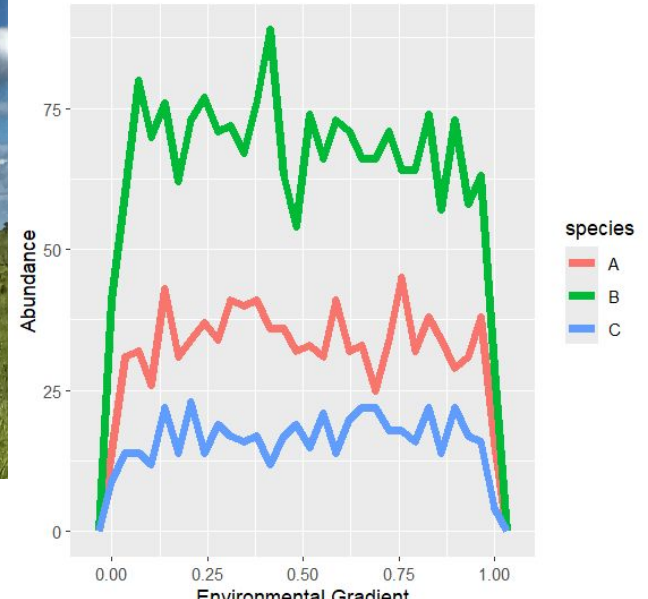
- E.g., treefall gaps in forests

These smaller-scale disturbances can increase species diversity

- Think about the spatial variation stuff we've discussed

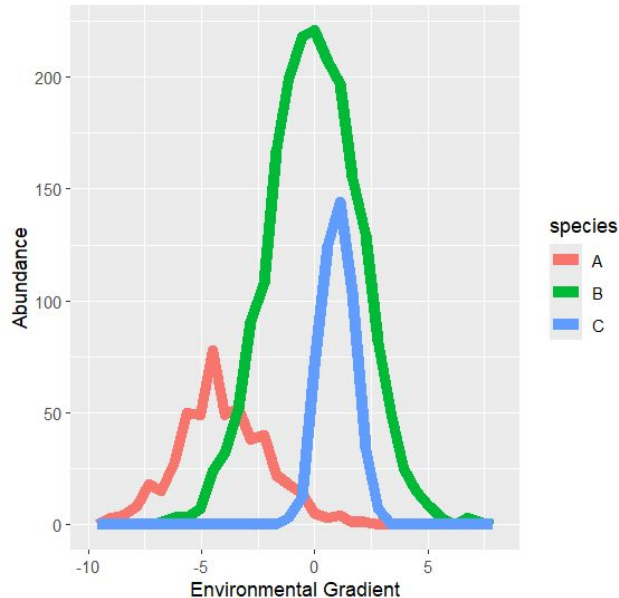
Succession

- Climax community assumes a set of species will occur at a given environment
- Succession is just a return to this set



Succession

- The idea of fixed types of communities has been abandoned
- Communities as aggregations of individual species
- Species dominate because they're suited to the current environment

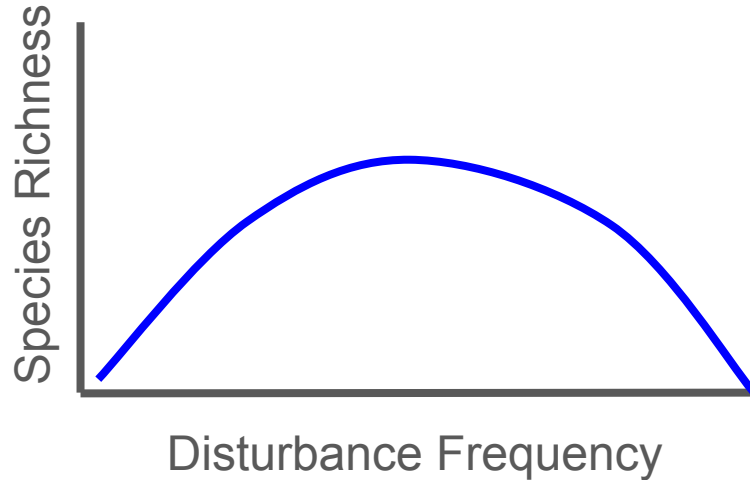


Succession

- Climax communities abandoned
- However, there is still a lot of study in how ecosystems change
- The focus now has shifted to community assembly, metacommunity, etc.

Intermediate Disturbance Hypothesis

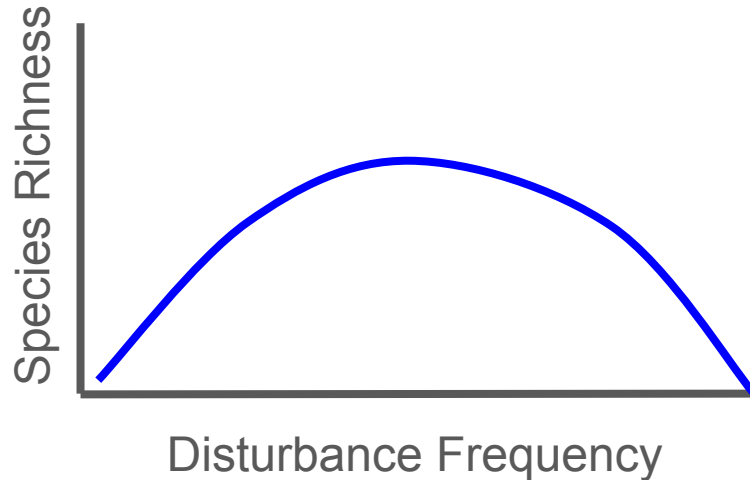
- species diversity highest at an intermediate level of disturbance



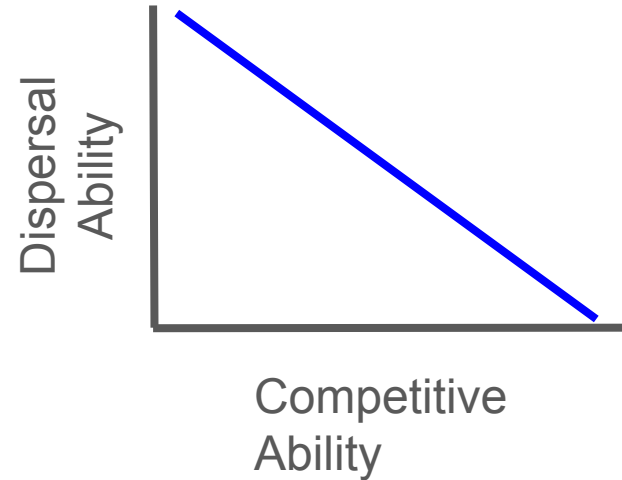
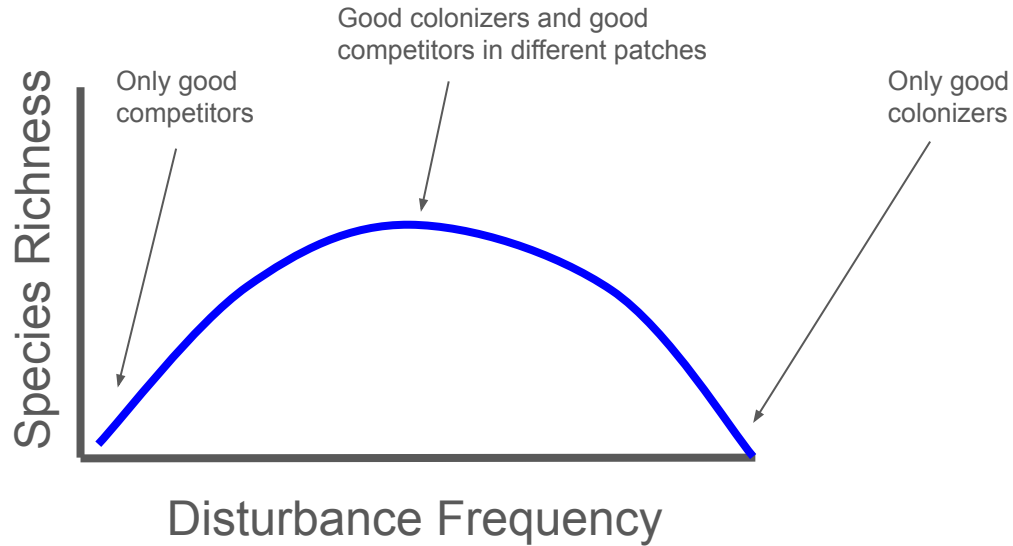
Intermediate Disturbance Hypothesis

- species diversity highest at an intermediate level of disturbance

What factors might drive this?



Intermediate Disturbance Hypothesis



Disturbance and Coexistence

- You may read or hear someone say something like:
 - “Disturbance promotes coexistence by reducing competition”
- This is NOT TRUE!
- Coexistence is only promoted when species are impacted differently
 - Needs to increase intraspecific competition relative to interspecific competition

MECHANISMS OF MAINTENANCE OF SPECIES DIVERSITY

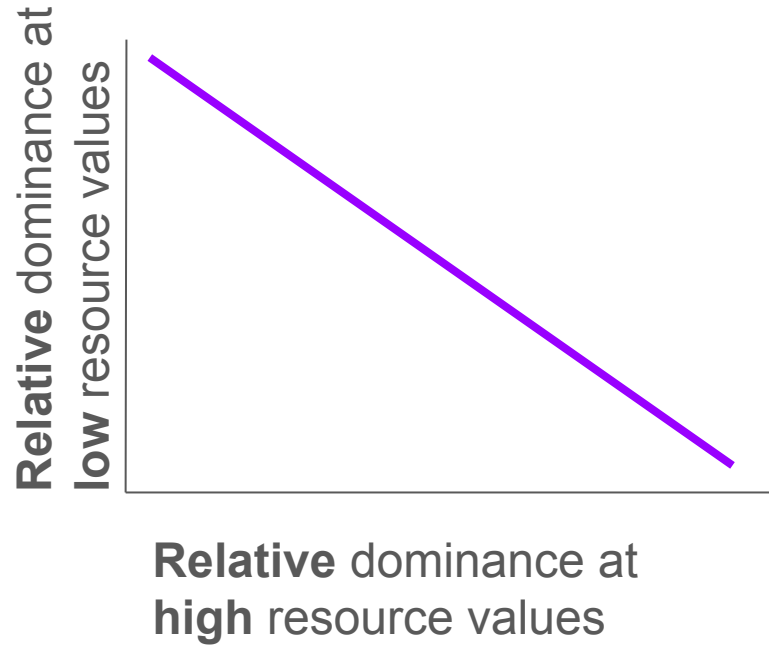
Peter Chesson

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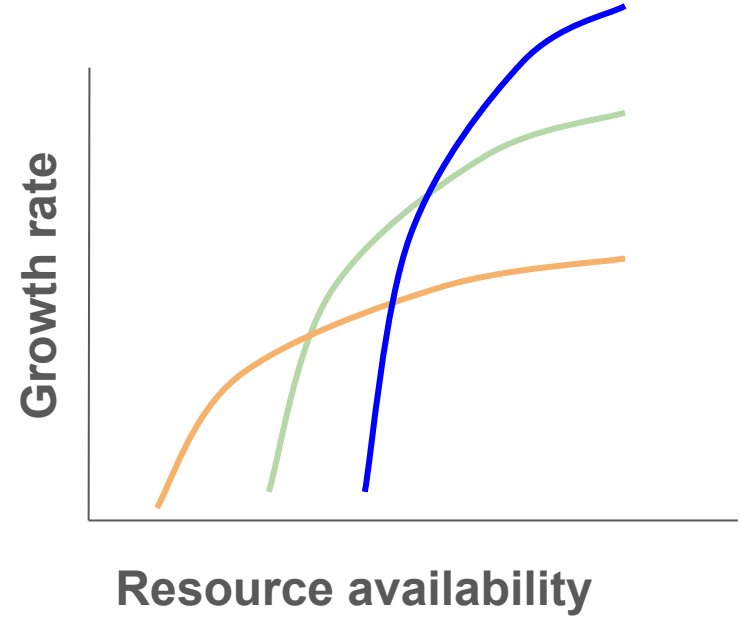
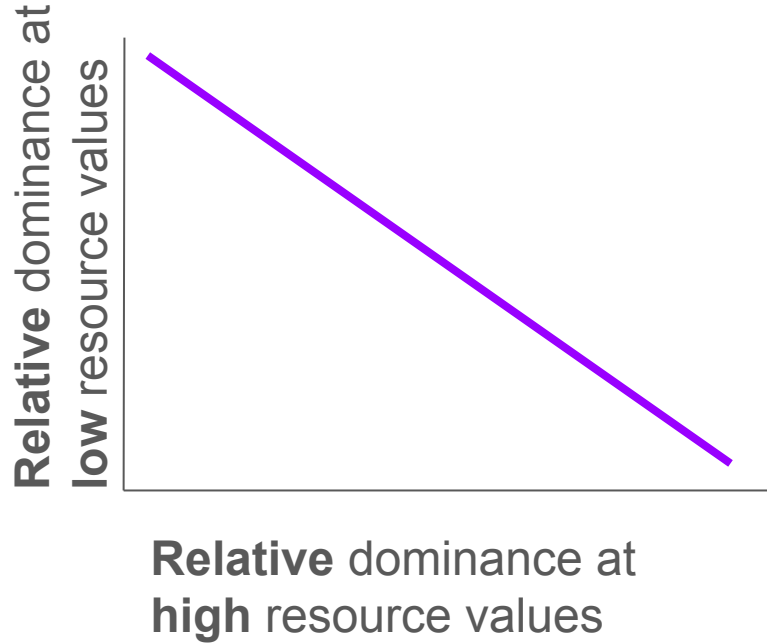
Temporal Variation and Coexistence

- Temporal and spatial variation can interact to allow coexistence
- Other fluctuation-dependent coexistence mechanisms:
 - Storage effect
 - Relative non-linearity

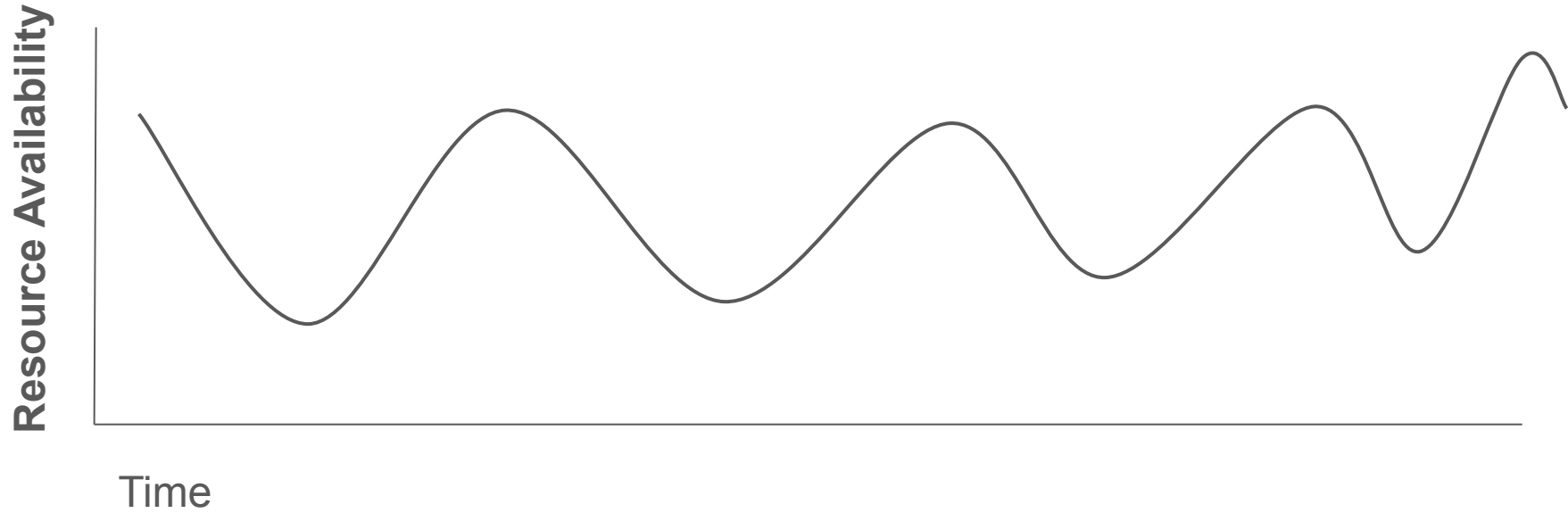
Relative Nonlinearity of competition



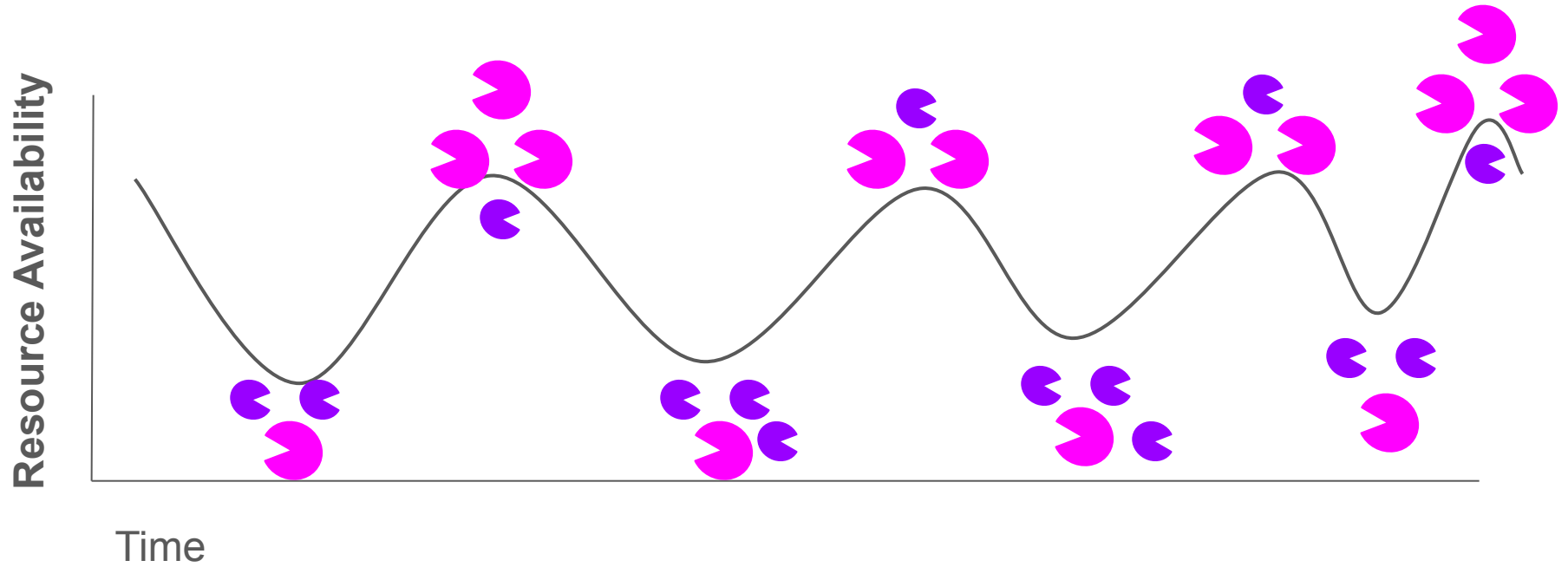
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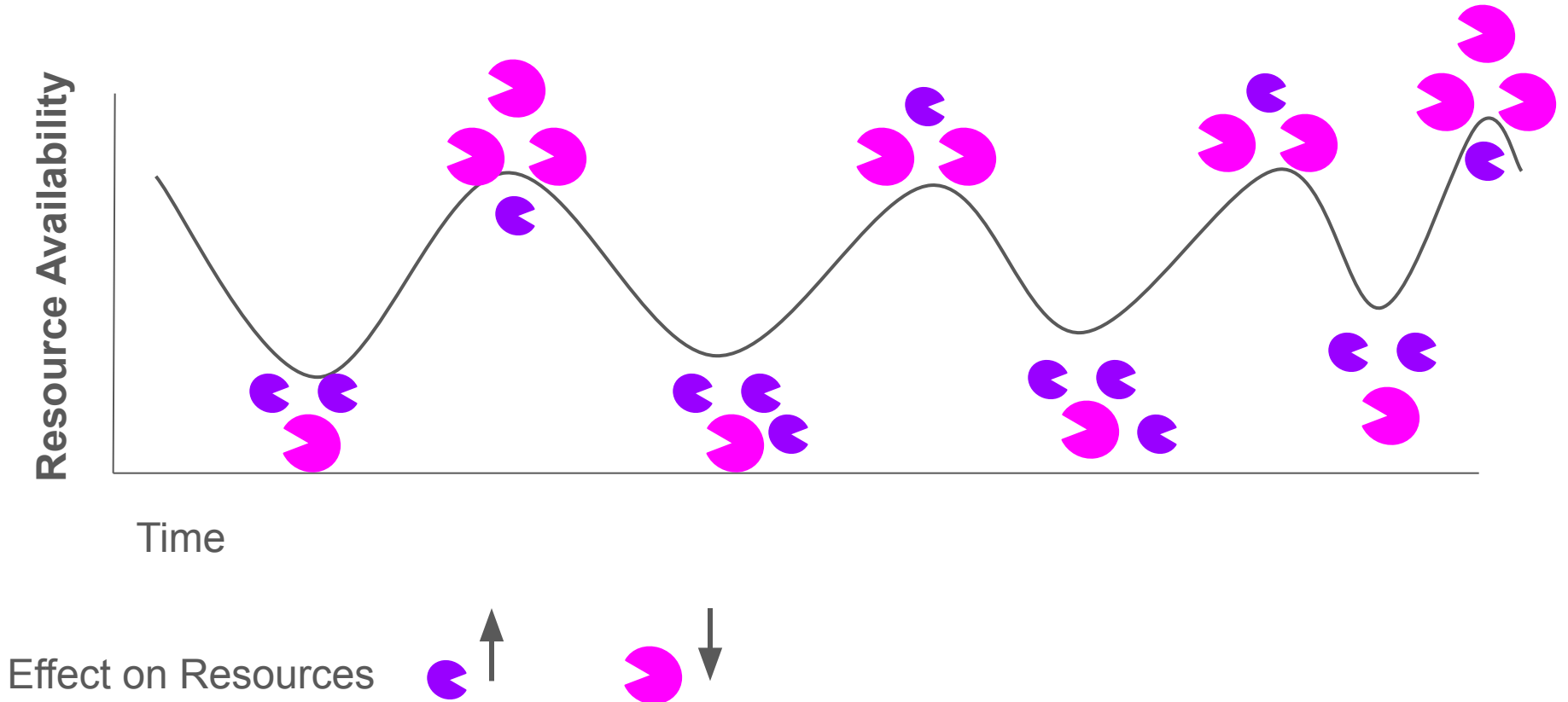
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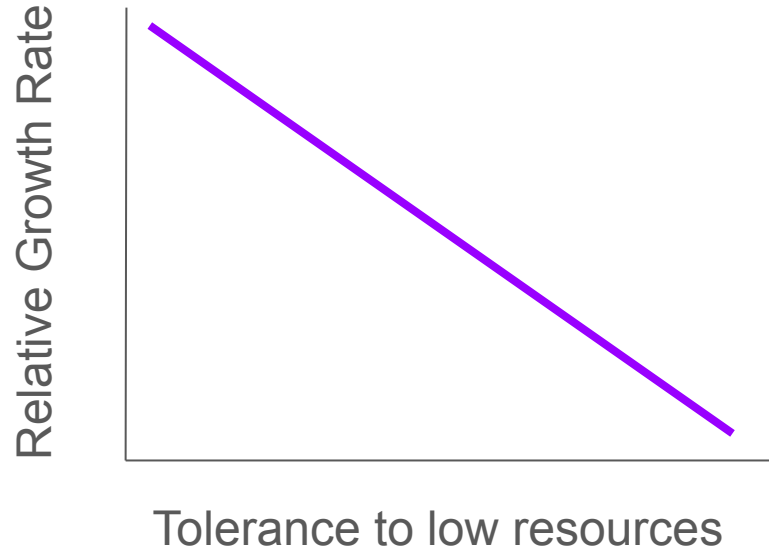


Storage Effect

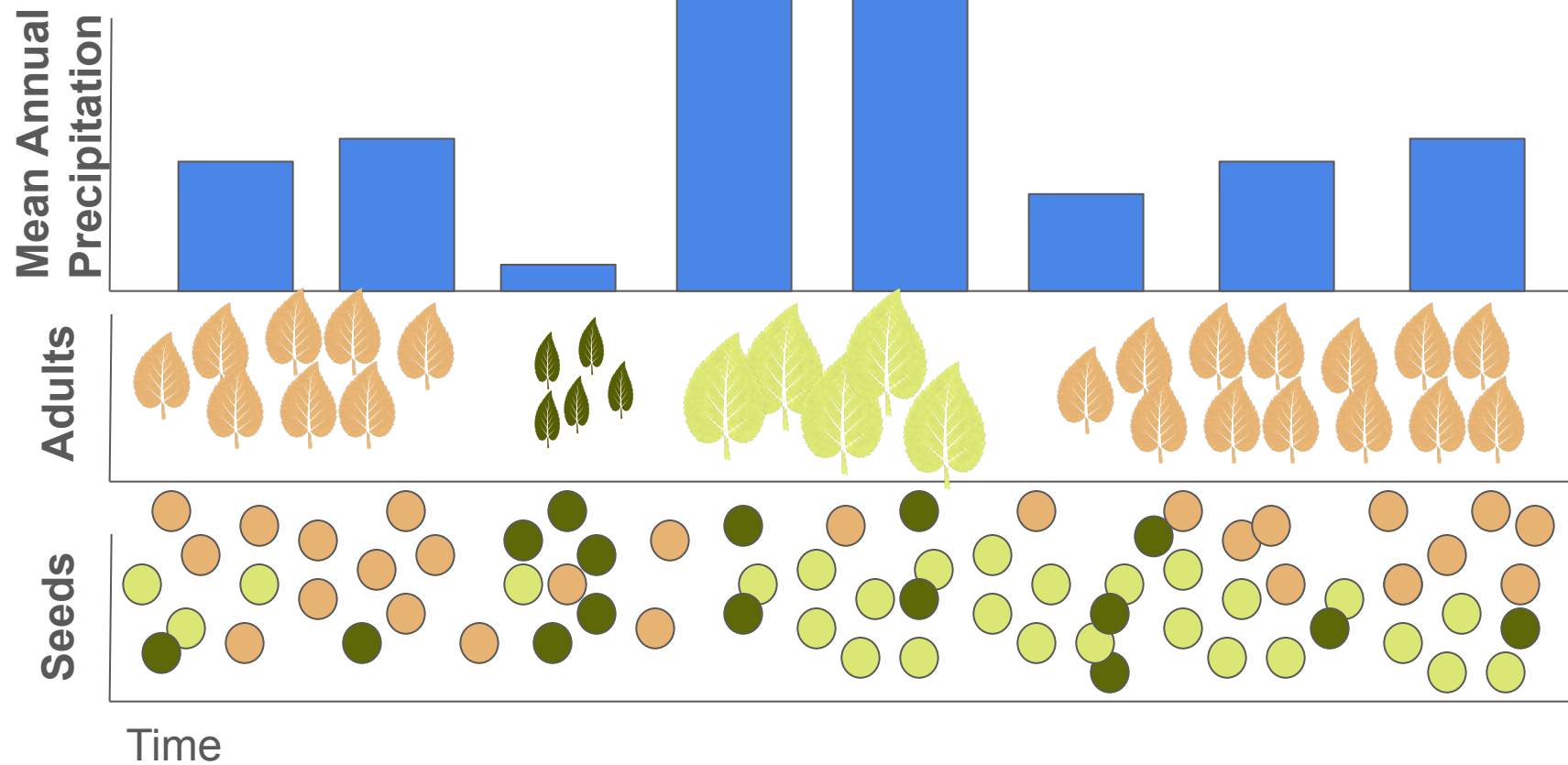
Storage Effect

- Species differ in the environment they do best in
- BUT they can persist through sub-optimal conditions
- (Note that there is a spatial version of this as well)

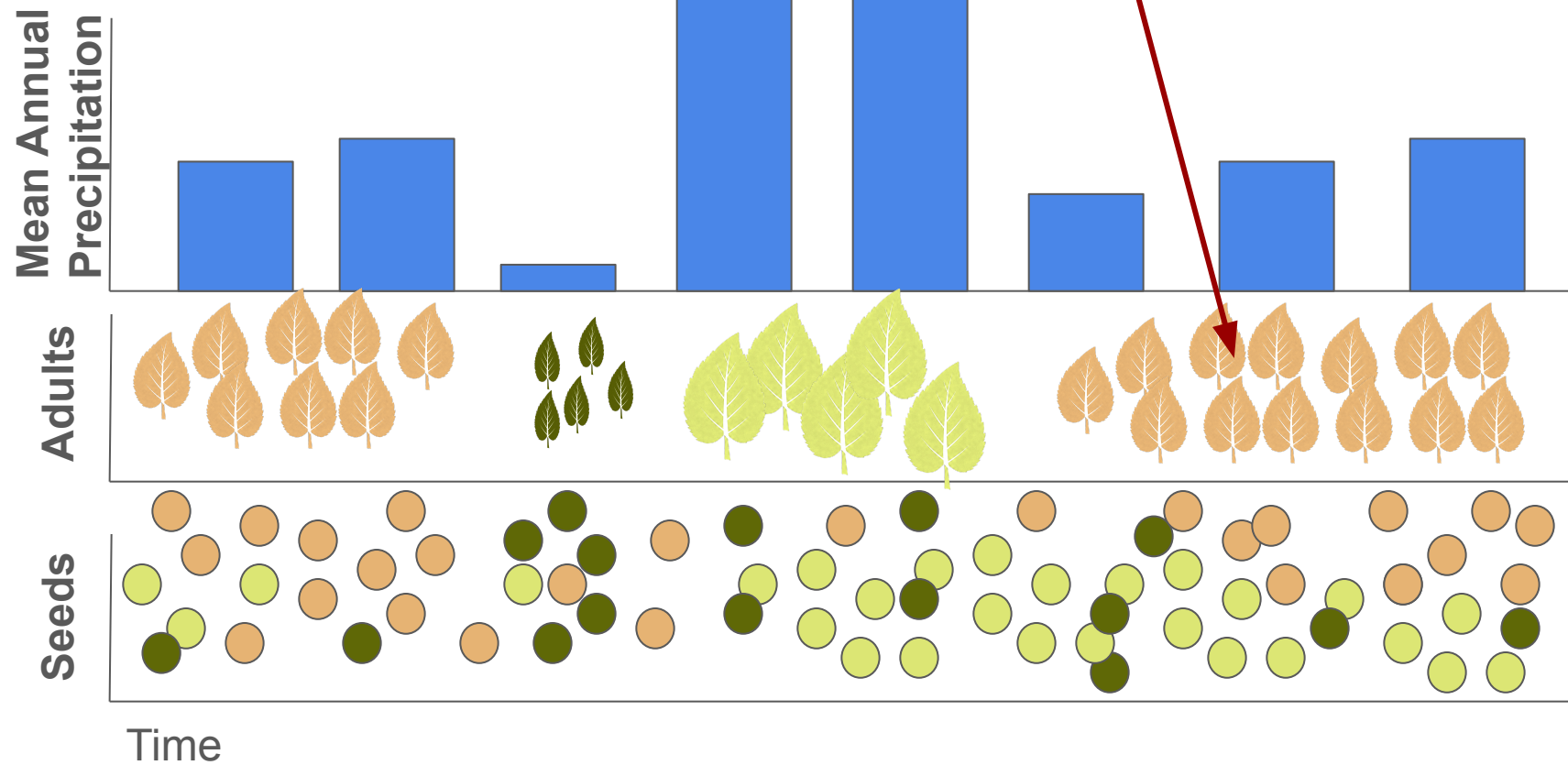
Storage Effect



Storage Effect



Storage Effect



Storage Effect

PNAS

Functional tradeoffs determine species coexistence via the storage effect

Amy L. Angert^{a,b,1}, Travis E. Huxman^{b,c}, Peter Chesson^b, and D. Lawrence Venable^b

^aDepartment of Biology, Colorado State University, Fort Collins, CO, 80523; and ^bDepartment of Ecology and Evolutionary Biology and ^cB2 Earthscience, University of Arizona, Tucson, AZ 85721

Communicated by José Sarukhán, Universidad Nacional Autónoma de México, Mexico D.F., Mexico, April 28, 2009 (received for review March 23, 2009)

How biological diversity is generated and maintained is a fundamental question in ecology. Ecologists have delineated many demographic decoupling of species arises from partially uncoupled responses to environmental variation, (i) the strength

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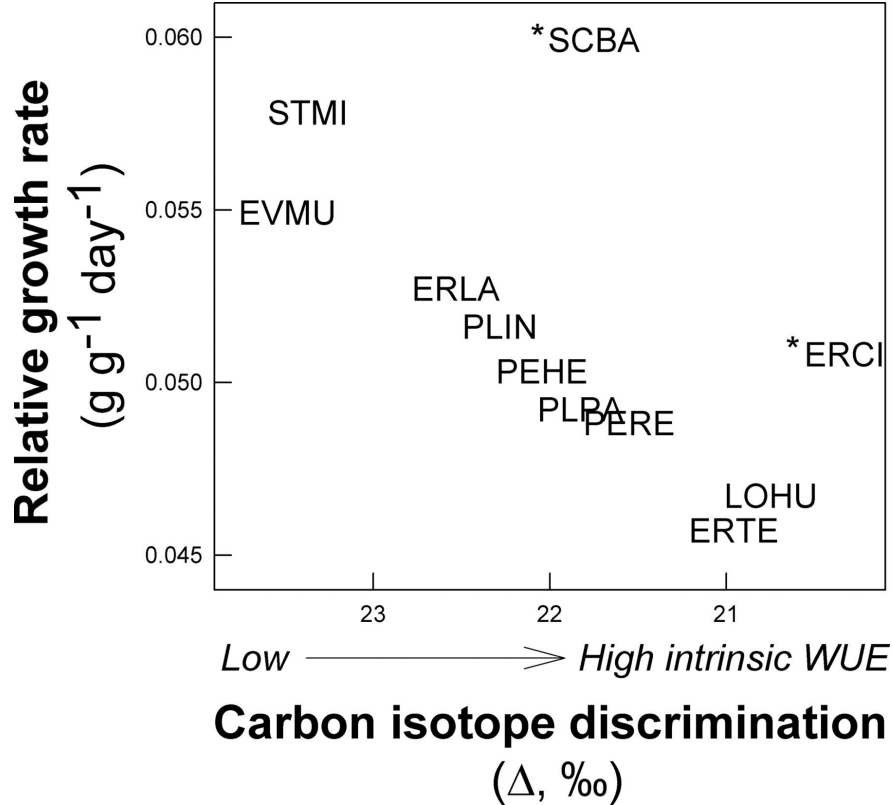
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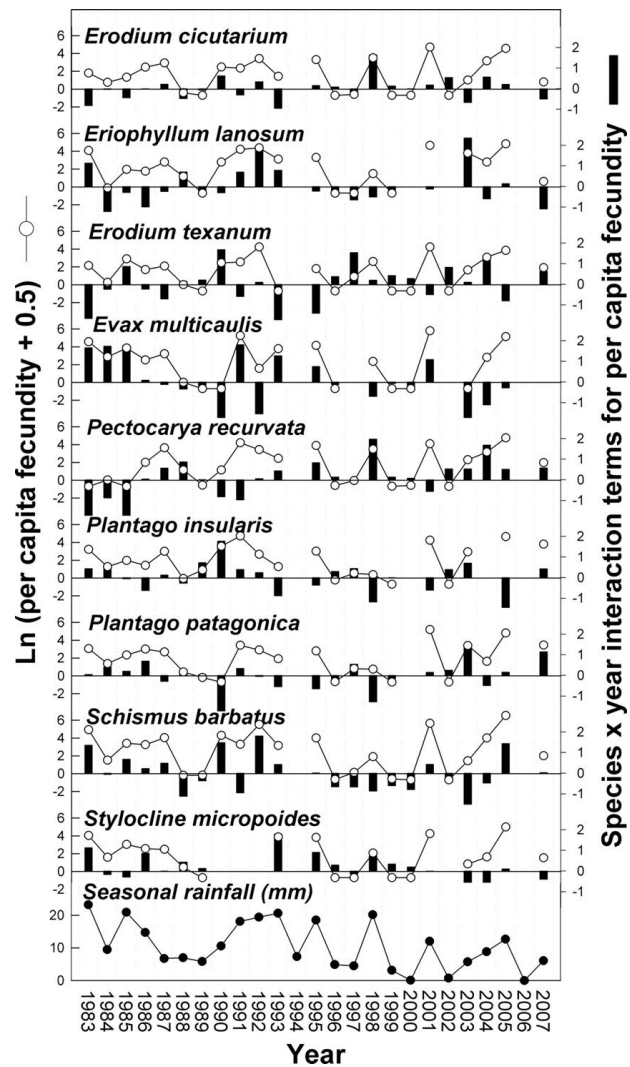
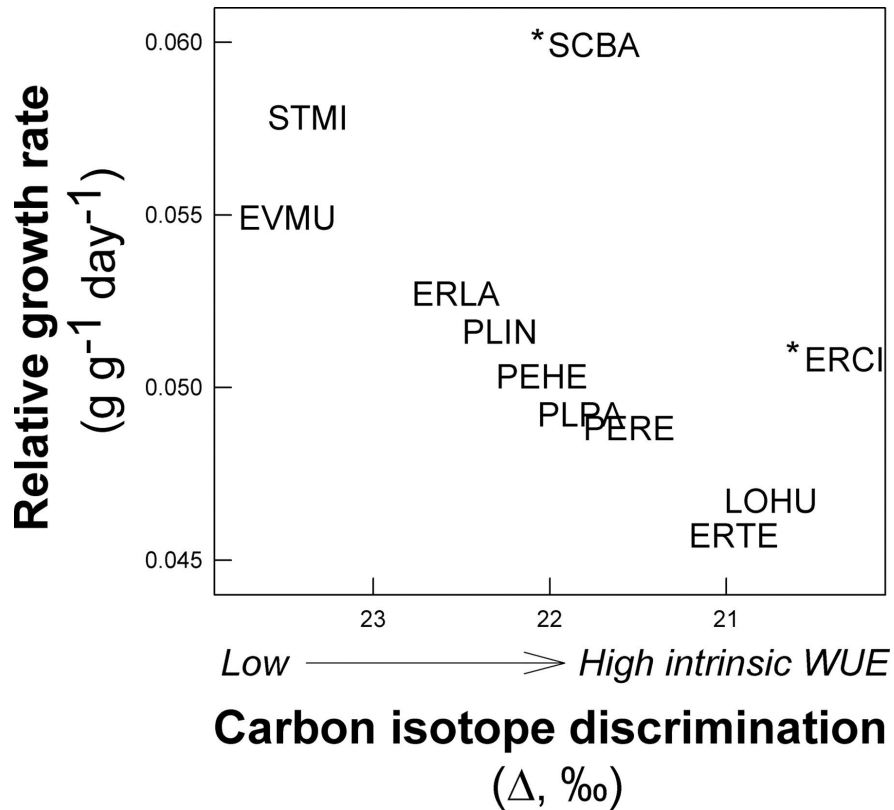
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Storage Effect



Storage Effect



Storage Effect

PNAS

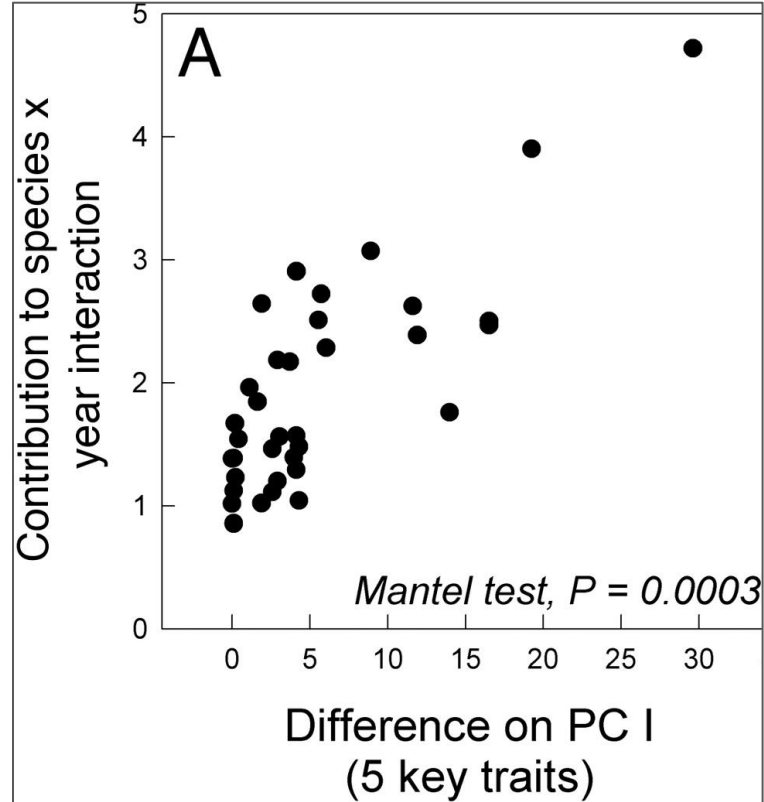
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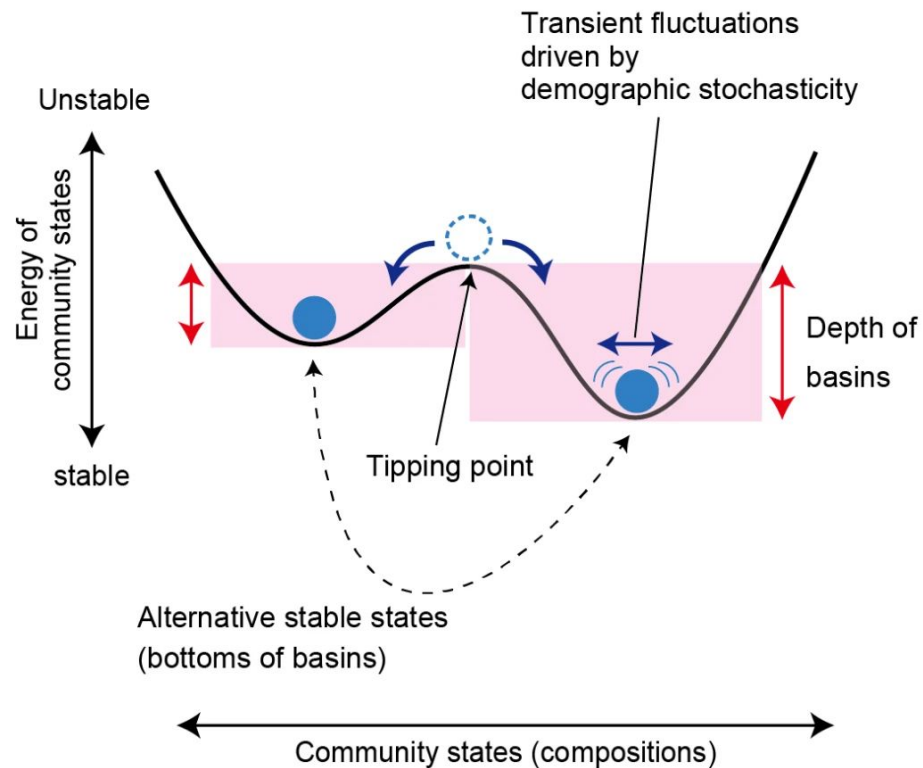
Regime Shifts and Alternative Stable States

Alternative Stable States

- Different communities, same environment
- Perturbations may cause a shift from one to the other

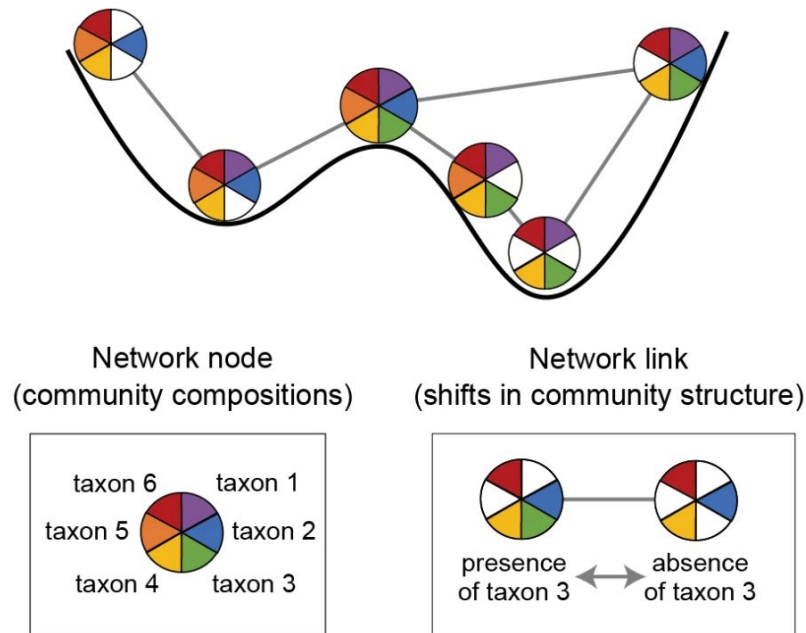
Regime Shifts and Alternative Stable States

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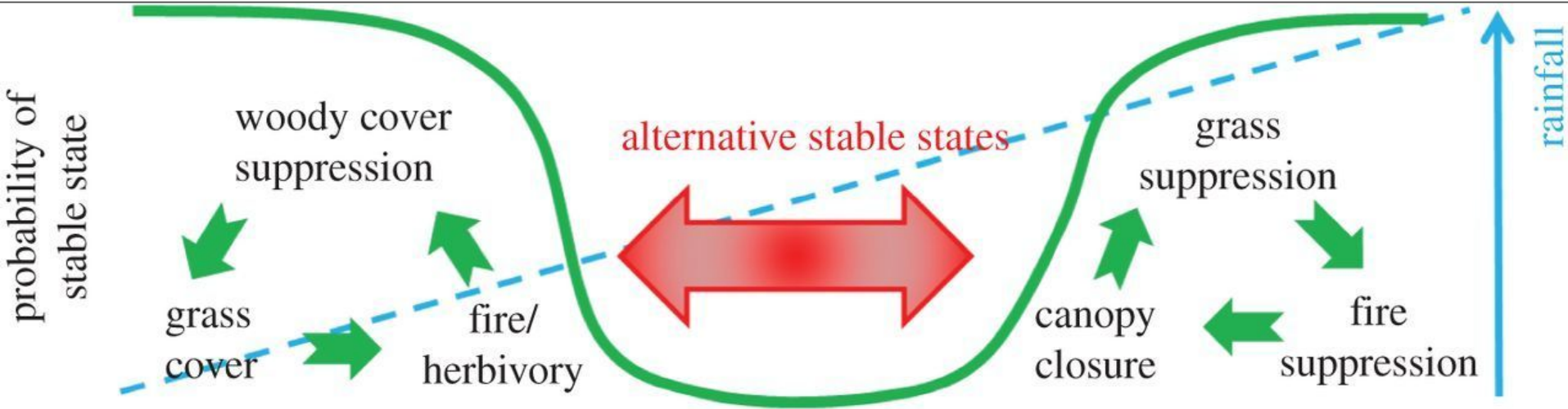


B

“Assembly graph”



Regime Shifts and Alternative Stable States



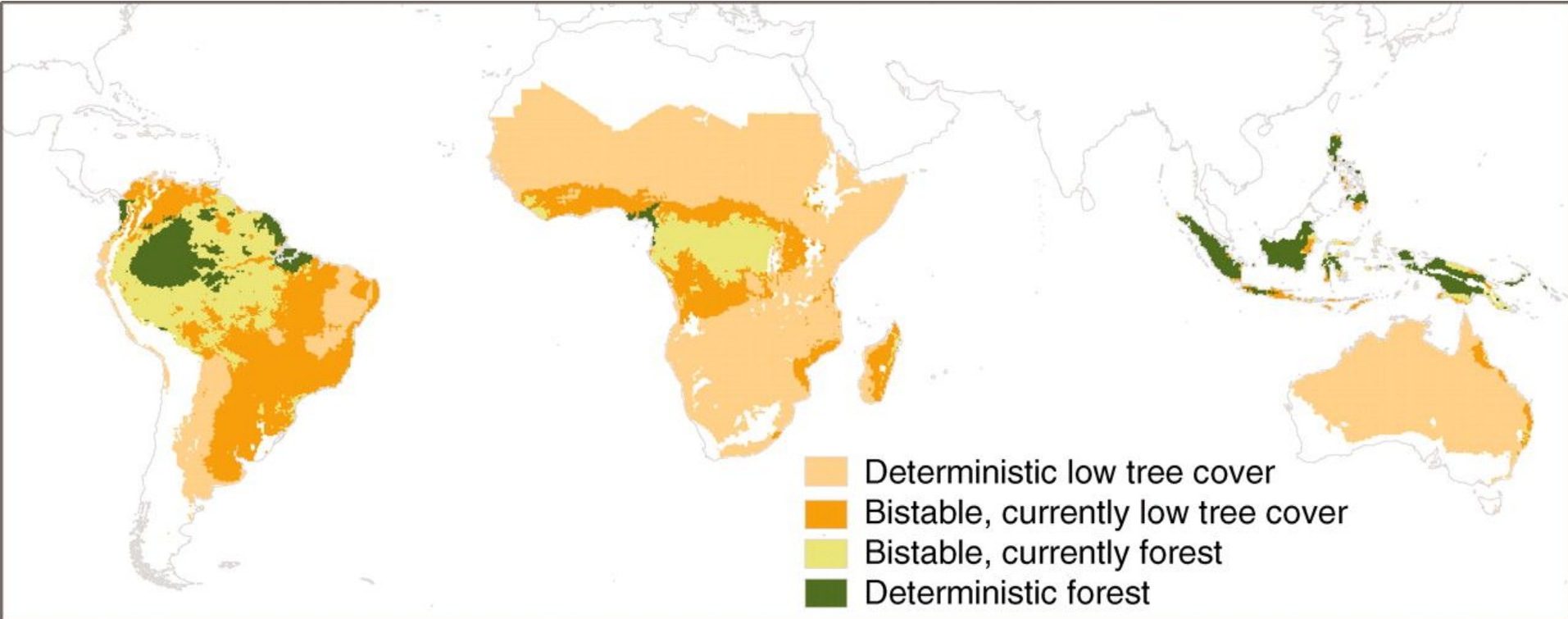
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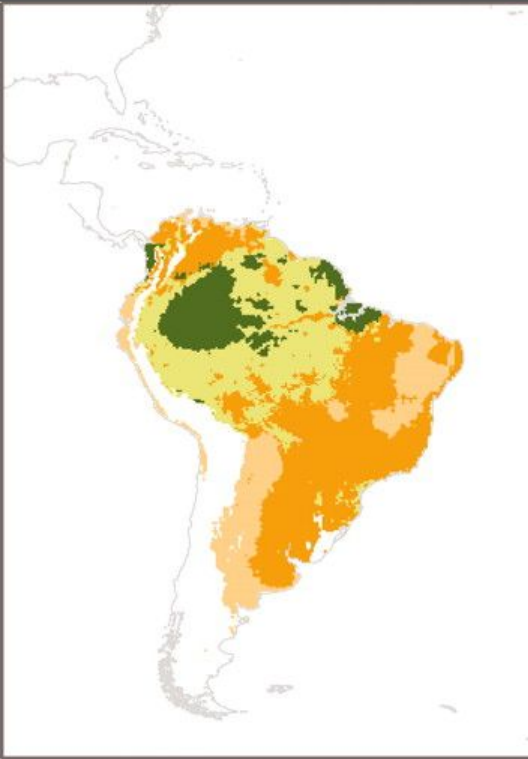
Many shades of green: the dynamic
tropical forest–savannah transition zones

Immaculada Oliveras[†] and Yadvinder Malhi[†]

Regime Shifts and Alternative Stable States



Regime Shifts and Alternative Stable States



A switch from forest to savannah has big implications

- Different species
- Different carbon sequestration
- Impacts on global climate models

Regime Shifts and Alternative Stable States

Will shifts be gradual or abrupt?

- Gradual gives us time to act
- Abrupt may leave us scrambling

Timing of change

Approach:

When will species be exposed to novel conditions?

Compared species' historical climates to projected future climates

Asked when species will be exposed to new climates

- Novel isn't necessarily bad
- But it does mean we don't KNOW a species can survive it
- High sensitivity, low specificity
- Make few assumptions

Abrupt expansion of climate change risks for species globally

Received: 23 May 2022

Alex L. Pigot¹, Cory Merow², Adam Wilson³ & Christopher H. Trisos⁴

Accepted: 14 April 2023

Timing of change

Abrupt onsets of exposure

nature ecology & evolution

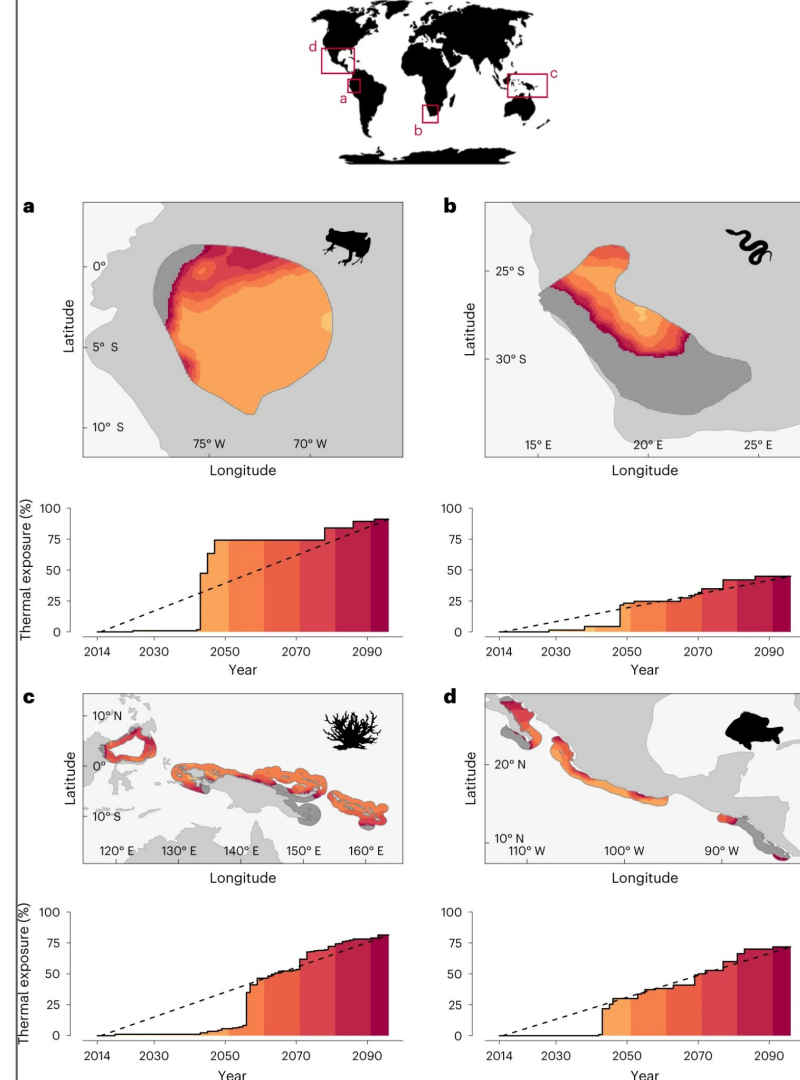
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Timing of change

Abrupt onsets of exposure

- Driven by realized niches

nature ecology & evolution

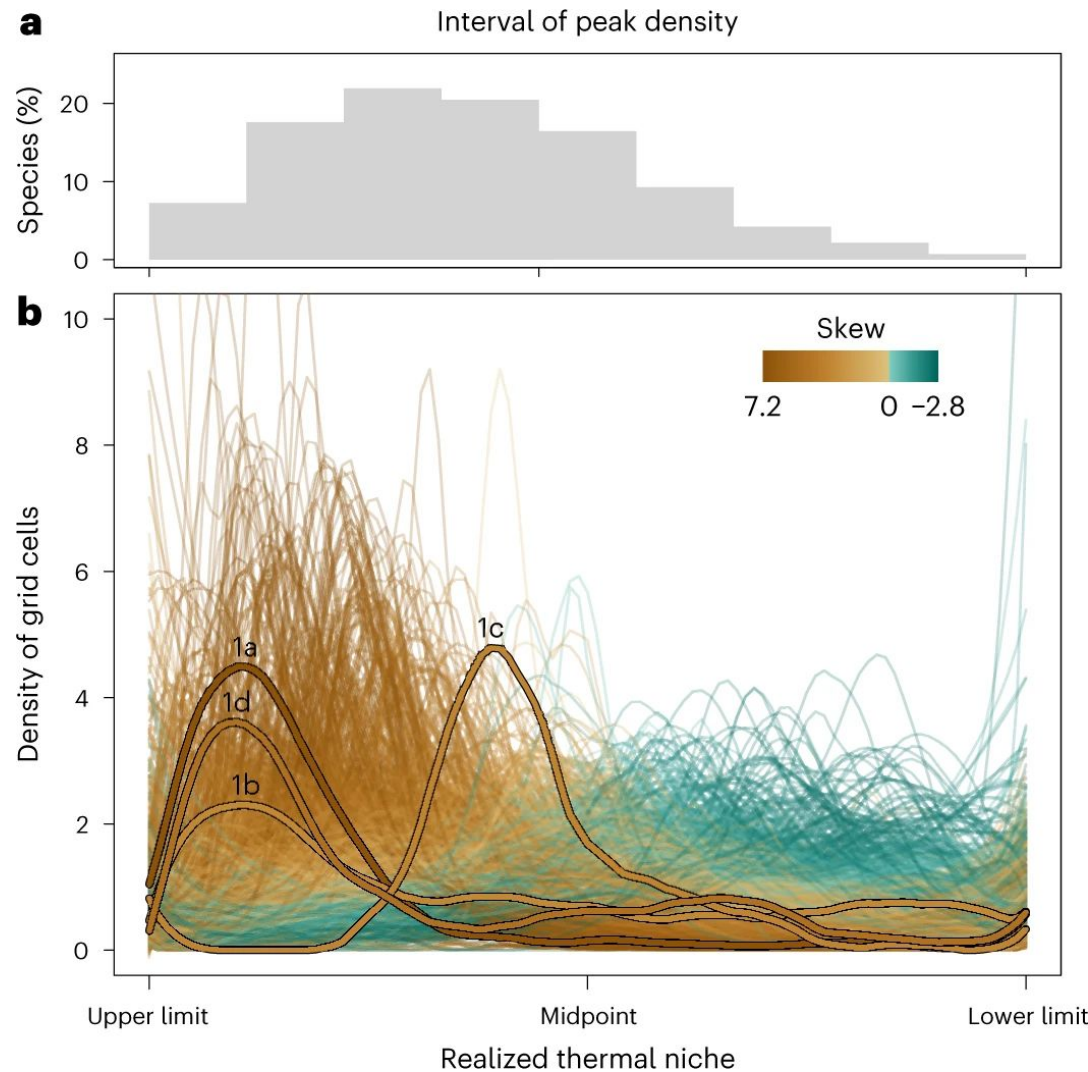
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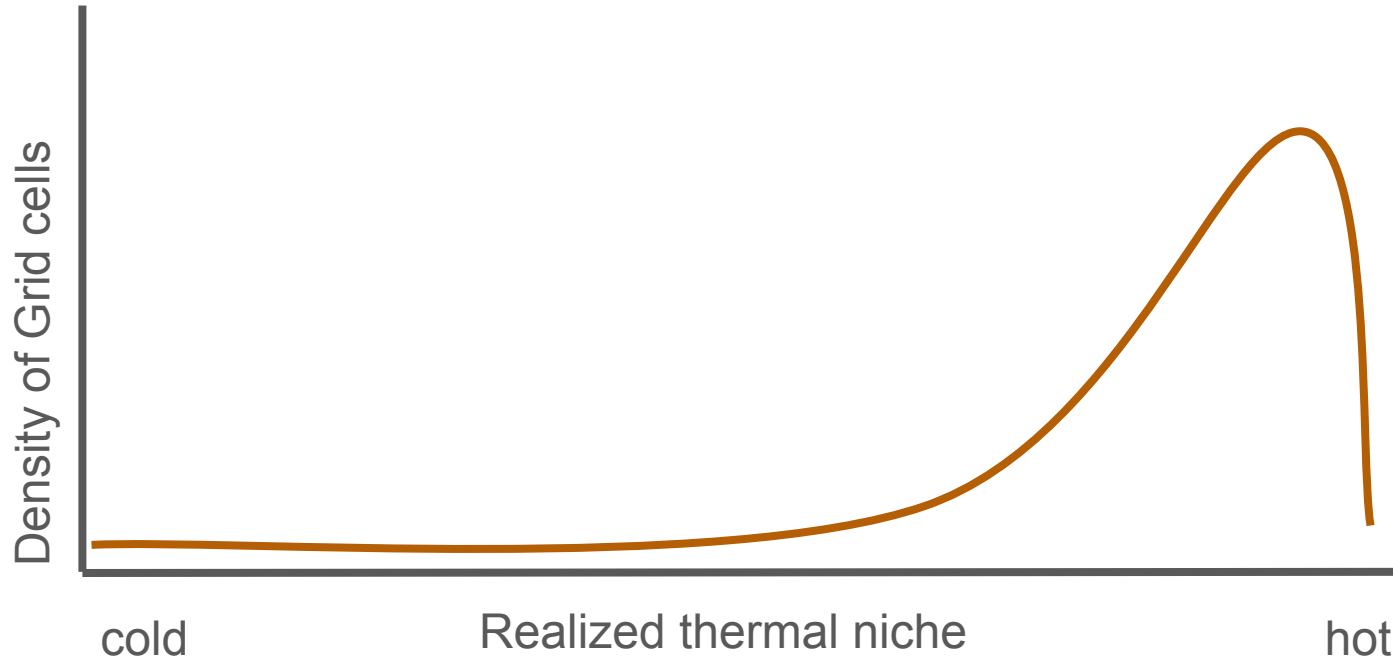
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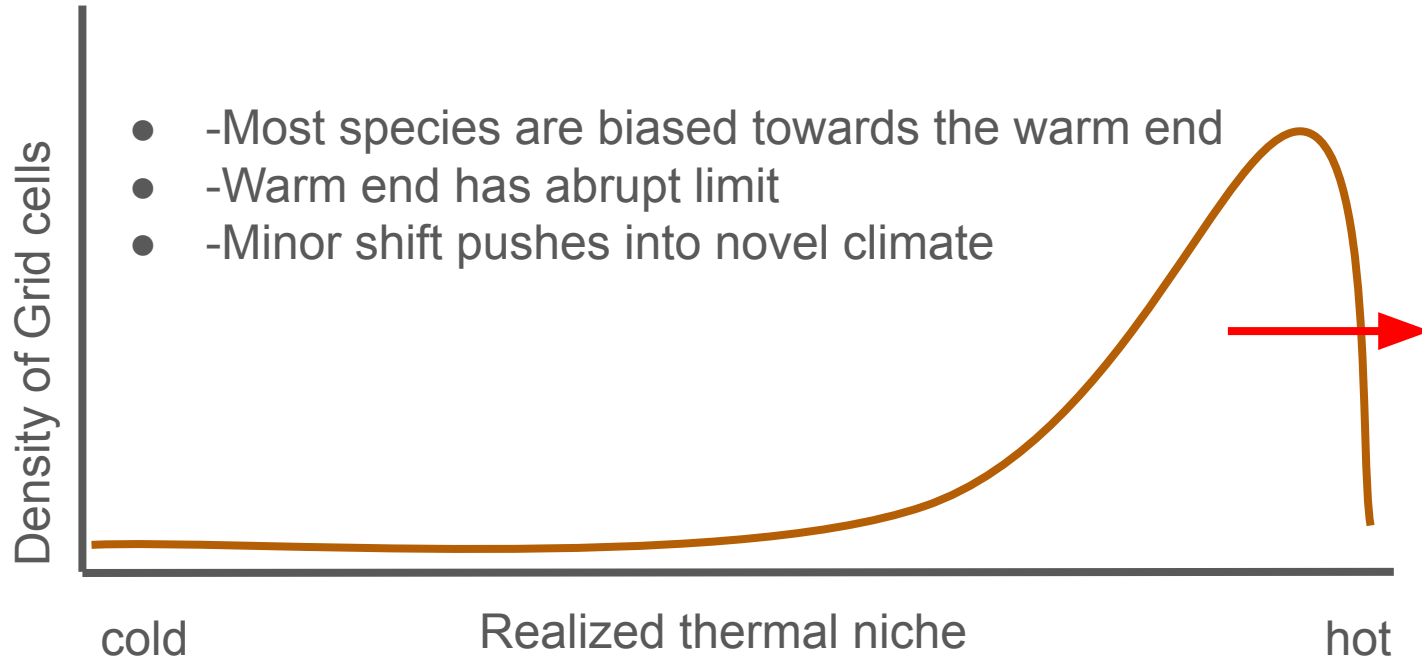


Timing of change

Note: x-axis is reversed from previous



Timing of change



Timing of change

Relatively slow increases in climate may cause sudden risks to species

- Need ways of identifying when exposure to new climate is a risk
- In 2023 and 2024, many species were exposed to novel temperatures
 - Need to understand which species were impacted and which weren't

Key ideas

- Succession
 - Intermediate disturbance hypothesis
- Temporal variation and coexistence
- Alternative Stable States
- Slow change but rapid exposure to new conditions

Next class: Students present something!